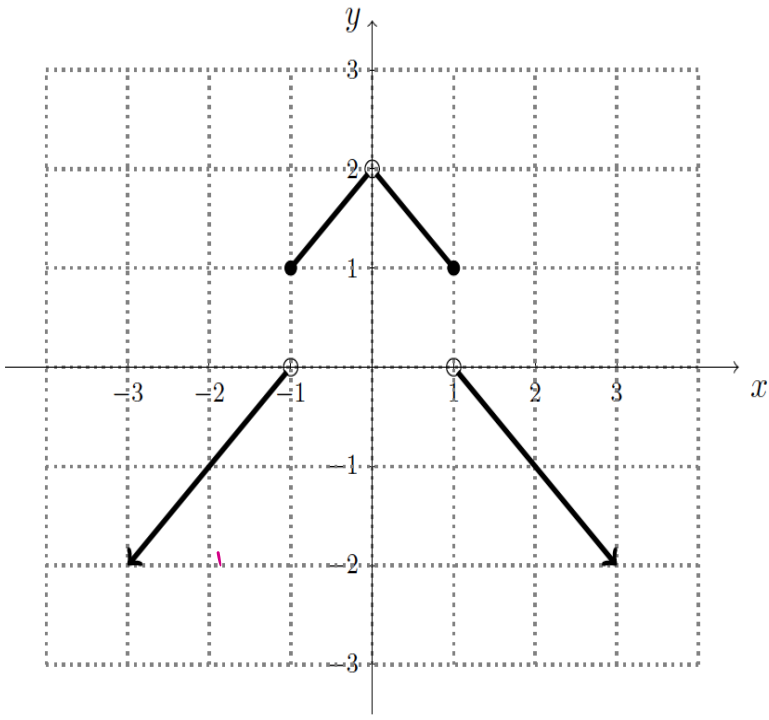
Answer All Questions

Question: 1

From the graph $y = f(x)$, answer the following:

a. $\lim_{x \to -1^+} f(x) =$ 1



b. $3f(-2) + f(2) =$

$3(-1) + (-1) = -4$

c. Is the function continuous at $x = 0$?

$\because$ $f(0)$ is not exist

$\therefore$ function is Not cont....

d. If $\lim_{x \to k^+} f(x) = 0$, then $k =$

Question: 2

a) The function $f(x) = \begin{cases} 2x - 3, & if\ x \neq a \\ 5, & if\ x = a \end{cases}$ is continuous at $x = a$. Find the value of a

$L = R = f(x)$

$2x - 3 = 5$

$2x = 8$

$\boxed{x = 4}$

b) Find the intervals in which $f(x) = \dfrac{\sqrt{6 - 3x}}{2}$ is continuous?

$6 - 3x \geq$ Zero

$\dfrac{-3x}{-3} \geq \dfrac{-6}{-3}$

$x \leq 2$

or

$6 - 3x \geq 0$

$\dfrac{6}{3} \geq \dfrac{3x}{3}$

$2 \geq x$

Question: 3

a. Use the functional value table to evaluate $\lim_{x \to -2} \sqrt{6-5x}$

LHR		RHR	
x	$f(x)$	x	$f(x)$
-2.1		-1.9	
-2.01		-1.99	
-2.001		-1.999	

b. Evaluate $\lim_{x \to 3} \frac{\sqrt{x}-\sqrt{3}}{x^3-3x^2}$

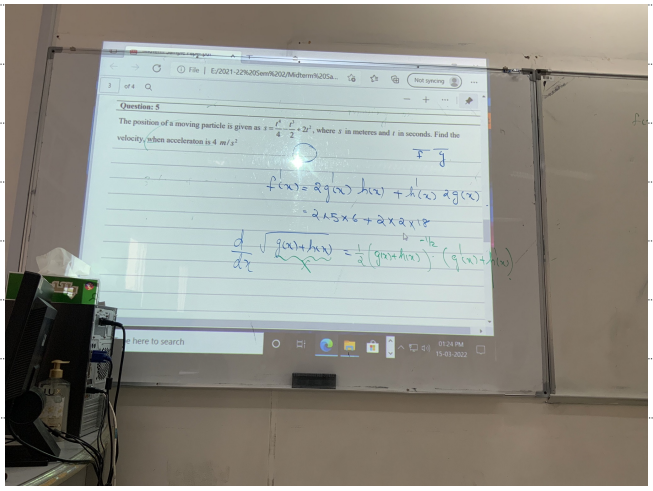
$$\frac{\sqrt{x}-\sqrt{3}}{x^2(x-3)} \cdot \frac{\sqrt{x}+\sqrt{3}}{\sqrt{x}+\sqrt{3}}$$

Question: 4

a) Use the definition to find derivative of $f(x)=\frac{2}{x+3}$ at $x=1$

$$\frac{f(x+h)-f(x)}{h}$$

b) Suppose that g and h are functions of x that are differentiable at $x=3$. Let $f(x)=2g(x)h(x)$ where $g(3)=18$, $h(3)=6$, $g'(3)=5$ and $h'(3)=2$. Find $f'(3)$



Question: 5

The position of a moving particle is given as $s=\frac{t^4}{4}-\frac{t^3}{2}+2t^2$, where s in meteres and t in seconds. Find the velocity, when acceleraton is 4 m/s^2

s' s''

Note : 2.4

$s' = \frac{4t^3}{4} - \frac{3t^2}{2} + 4t$

$s'' = 3t^2 - \frac{\cancel{6}^{3}t}{\cancel{2}} - 4$

$3t^2 - 3t = 0$

$3t^2 -$

$t = 0, 1$

Question: 6

If $y=5\sqrt{x^3}+3x^2-4$, Find $\frac{d^2y}{dx^2}$

$5.\frac{1}{2}(x^3)^{-\frac{1}{2}}.3x^2 + 6x$

$5.\frac{1}{2}.\frac{1}{2}(x^3)^{-\frac{3}{2}}.3x^2.6x+6$

$\frac{5}{4}(x^3)^{-\frac{3}{2}}3x^2 6x + 6$

Question: 7

Find the derivative of the following:

a) $y = \frac{\pi x}{2} + \sin\sqrt{x^2 - 4}$

$\frac{\pi}{2} + \cos\sqrt{x^2-4} \cdot \frac{1}{2}(x^2-4)^{-1/2} \cdot 2x$

b) $y = e^{-z^2}\sqrt{\ln 2}$

$\sqrt{\ln 2} \quad e^{-z^2} \cdot -2z$

c) $y = \frac{5^x}{3\tan^{-1} x + 5}$ $\quad \frac{f}{g}$

$$\frac{5^x \ln 5 \cdot 3\tan^{-1} x + 5 - \left(3\frac{1}{1+x^2} \cdot 5^x\right)}{(3\tan^{-1} x + 5)^2}$$

Question:8

a) Find the equation of the tangent of the curve $y = \overbrace{\left(\frac{\pi}{6} - \theta\right)}^{f}\overbrace{\csc\theta}^{g}$ at $\theta = \frac{\pi}{6}$

$y' = -1\csc\theta + -\csc\theta\cot\theta \cdot \left(\frac{\pi}{6} - \theta\right)$

$x_1 = \frac{\pi}{6}$

$y_1 = \left[\frac{\pi}{6} - \frac{\pi}{6}\right] \leftarrow$ $y_1 =$ التعويض عن x_1 في المعادلة.

m_z